

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2018

Course: B. Tech. in Mechanical Engineering

Semester: III

Subject Name: Materials Science and Metallurgy

Subject Code: BTMEC 302

Date: 03/12/2018

Marks: 60

Duration: 3 Hrs.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is permitted.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve the following.		12
A) Derive the expressions for planar atomic densities on (100), (110), and (111) planes of both BCC and FCC crystal structures.	(Understand, CO1)	
B) Explain slip mechanism of plastic deformation w.r.t. conceptual meaning, its occurrence due to the movement of edge and screw dislocations, and comparison with twinning.	(Understand)	
Q.2 Solve the following.		12
A) A tension test was conducted on steel specimen of diameter 12.5 mm and gauge length 50 mm. The loads at lower and upper yield points were recorded to be 45000 and 46000 N respectively. The maximum and fracture loads were 75000 and 50000N respectively. The gauge length after fracture was 62.5 mm. The diameter at fracture was found to be 8 mm. At a load of 20000 N, the total extension was 0.035 mm. Determine the following: (i) Lower yield stress (ii) Ultimate tensile stress (iii) True fracture stress (iv) Percentage elongation (v) Percentage reduction in area (vi) Modulus of resilience.	(Apply, CO 2)	
B) Discuss Vickers Hardness Test w.r.t principle of working, indenter details, formula, advantages, and limitations.	(Understand)	
Q. 3 Solve the following.		12
A) Describe the types of solid solutions. Explain Hume-Rothery's rules of solid solubility.	(Remember)	
B) What is the importance of T-T-T diagrams? Explain the procedure to determine these diagrams with the help of schematic diagrams.	(Understand)	
Q.4 Solve Any Two of the following.		12

- A) Define heat treatment and give its objectives. Give the names of two different heat treatment processes along with the major objective of each. *(Understand)*
- B) Explain tempering process w.r.t. purposes, process details, types, and variation of properties with tempering temperature. *(Understand)*
- C) Discuss induction hardening process w.r.t. principle of working, process details, advantages, and limitations. *(Understand)*

Q. 5 Solve Any Two of the following.

12

- A) Describe the procedure of specimen preparation for microscopy. *(Understand)*
- B) Explain the principle of working of optical metallurgical microscope. Compare it with electron microscope. *(Remember)*
- C) Discuss sulphur print test w.r.t. purpose, significance, procedure, and chemical reactions. *(Understand)*

Q. 6 Solve the following.

12

- A) Explain dispersion strengthening w.r.t. basic mechanism, critical factors, advantage, and commercial examples. *(Remember)*
- B) Describe ultrasonic inspection technique w.r.t. principle of working, types, and applications. *(Understand)*

☆ End of Paper ☆

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – SUMMER 2019

Course: B. Tech in Mechanical Engineering

Sem: III

Subject Name: Material Science and Metallurgy

Subject Code: BTMEC302

Max Marks: 60

Date: 29-05-2019

Duration: 3 Hr.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Q. 1 Solve Any Two of the following.

Marks

12

- A) What is plastic deformation? Explain with neat sketch plastic deformation by slip.
- B) Define the term coordination number, packing density and atomic radius. And prove the packing density of the FCC unit cell is 0.74
- C) Classify crystal imperfections and explain screw dislocation in details.

Q.2 Solve Any Two of the following.

12

- A) Derive relation between engineering and true stress-strain and also draw true stress-strain curve for mild steel.
- B) Classify hardness tests. Explain Rockwell hardness test in detail.
- C) Explain with neat sketch Izod impact test. How Izod impact test is different from charpy impact test.

Q. 3 Solve Any two of the following.

12

- A) Explain Iron-Iron carbide equilibrium diagram with neat sketch.
- B) With neat sketch describe the mechanism of transformation of austenite to bainite.
- C) Define the critical cooling rate of a steel and show the critical cooling rate on a TTT diagram.

Q.4 Solve Any Two of the following.

12

- A) What is annealing? State the purpose of annealing and also plot the heating temperature band for full annealing.
- B) Define hardenability. Explain Jominy-End quench test for hardenability with neat sketch.
- C) Classify surface hardening processes. Explain any one in detail.

Q. 5 Solve Any two of the following.

12

- A) Explain the procedure followed in specimen preparation in metallography.
- B) Describe spark test. What observations to be noted and also draw spark patterns for low and high carbon steels

- C) State and explain working principle of metallurgical microscope with neat sketch.

Q. 6 Solve Any two of the following.

12

- A) Explain magnetic particle inspection and also enumerate the limitations of magnetic particle inspection.
- B) What are the various strengthening mechanisms? Explain any one method in detail.
- C) Explain the principles of the following methods of the inspection.
- 1). Dye Penetrant inspection
 - 2). Ultrasonic inspection.
 - 3). Eddy current testing,

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103
Winter Semester Examination -- Dec. 2019

Branch: B Tech Mechanical Engineering

Sem.:- III

Subject:- Material Science and Metallurgy (BTMEC302)

Marks: 60

Date:- 12/12/2019

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

	(Marks)
Q.1.	(12)
a) Explain in short different imperfections in crystal structures	6
b) Draw stress strain diagram for mild steel, show effect of Carbon content on shape of stress strain diagram	6
Q2.	(12)
a) What is solid solution? Differentiate between substitutional and interstitial solid solution?	6
b) Draw Fe-Fe ₃ C equilibrium diagram. Show all temperatures and phases	6
Q3.	(12)
a) Draw neat labeled TTT diagram for eutectoid steel and give stepwise experimental procedure for drawing it.	6
b) What is annealing? List different types of annealing along with their purpose.	6
Q4.	(12)
a) Explain with neat sketches different types of flame hardening.	6
b) What is surface hardening? Explain induction hardening with neat sketch.	6
Q5.	(12)
a) Explain steps in specimen preparation for microscopy.	6
b) For spark test draw the sparks for the following specimen 1. CI 2. MS 3. HSS	6
Q6.	(12)
a) Write a note on strain hardening	6
b) Explain dye penetrant test. What is its application?	6

Paper End

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. (Second Year) Branch : Mechanical Semester : III Subject Code & Name: BTMEC302 Materials Science and Metallurgy Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		4
A)	Draw the crystal Structure of FCC and HCP.	CO1	2
B)	Define toughness and hardness.	CO2	2
C)	What is Gibbs phase rule?	CO3	2
Q.2	Solve Any Two of the following.		16
A)	Explain screw dislocation and edge dislocation in detail.	CO1	8
B)	What is importance of equilibrium diagram? Explain in detail.	CO3	8
C)	What are the different methods of hardness measurement? Explain any two.	CO2	8
Q. 3	Solve Any Two of the following.		16
A)	Draw the neat sketch Iron-iron carbide equilibrium diagram and explain phases existing on it.	CO3	8
B)	Explain hardening heat treatment process. Why tempering is only followed by hardening process?	CO4	8
C)	Classify surface hardening processes. Explain any two methods of hardening.	CO4	8
Q.4	Solve Any Two of the following.		16
A)	Write down short note on Sulphur printing test and Spark test.	CO5	8
B)	What are the types of Strengthening Mechanisms? Explain any one in detail.	CO6	8
C)	Write about the specimen preparation procedure in detail.	CO5	8
Q. 5	Solve Any Two of the following.		8
A)	Differentiate between annealing and normalizing processes.	CO3	4
B)	What are the steps involved in dye Penetrant inspection?	CO5	4
C)	What is objective of TTT diagram?	CO4	4
*** End ***			

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination: 2022 - 23 Program: B. Tech. Branch: Mechanical Engineering Semester: III Course Code & Name: BTMEC302 Materials Science and Metallurgy Max. Marks: 60 Date: 15.3.2023 Duration: 3 Hrs.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculator is permitted. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What is atomic packing factor? Prove that the atomic packing factor for FCC structure is 0.74.	Understand	6
B)	What are the mechanisms of plastic deformation? Explain slip mechanism due to the movement of edge dislocations and screw dislocations with neat sketch.	Understand	6
C)	Discuss Vickers hardness test w.r.t working principle, indenter details, formula, advantages and limitations.	Remember	6
Q.2	Solve Any Two of the following.		12
A)	Draw Iron-Carbide Equilibrium diagram and define all the phases.	Remember	6
B)	Explain Hume-Rothery's rules of solid solubility. What is Gibb's phase rule.	Remember	6
C)	What is the importance of TTT diagram? Explain the procedure to determine these diagrams with the help of neat sketch.	Understand	6
Q. 3	Solve Any Two of the following.		12
A)	What are the objectives of heat treatment? Explain different types of annealing processes with the help of schematic diagrams.	Understand	6
B)	Define hardenability and explain, in detail, the Jominy End Quench Test with neat sketch.	Remember	6
C)	Explain induction hardening process in detail. Give its advantages and limitations.	Understand	6
Q.4	Solve Any Two of the following.		12
A)	Describe various steps in specimen preparation for microscopy.	Understand	6
B)	Explain the construction and working principle of metallurgical microscope with neat sketch.	Remember	6
C)	Discuss Spark test in detail and draw the spark pattern for the Mild Steel and High Carbon Steel.	Understand	6

Q. 5	Solve <i>Any Two</i> of the following.		12
A)	Describe Magnetic Particle testing w.r.t. principle of working and applications.	Understand	6
B)	Explain Dye Penetrant Test in detail w.r.t. basic principle, steps, and limitations.	Understand	6
C)	Explain basic mechanism of dispersion strengthening. What are its critical factors, advantages and applications?	Remember	6
	*** End ***		

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

Course: S.Y. B. Tech. Branch: Mechanical Engineering Semester: III

Subject Code & Name: BTMES304 Material Science and Metallurgy

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) Write Classification of Engineering materials.	CO1	06
B) Explain Imperfection in crystal Structure.	CO1	06
C) What are the types of impact test? Explain the Procedure of Charpy Impact test.	CO1	06
Q.2 Solve Any Two of the following.		
A) What is a TTT diagram? What is its significance and use?	CO2	06
B) Classification of steel on the basis of location on Fe-Fe ₃ C equilibrium diagram (Hypo and Hyper eutectoid steels)	CO2	06
C) Write AISI Specification for steels.	CO2	06
Q. 3 Solve Any Two of the following.		
A) Explain the various steps of the heat treatment and the purpose of each step.	CO3	06
B) Which type of cooling media used in heat treatment list them and explain.	CO3	06
C) Write short note on Quenching and Hardenability	CO3	06
Q.4 Solve Any Two of the following.		
A) What is microscopy? Write down the steps of specimen preparation.	CO4	06
B) Explain the procedure of spark test.	CO4	06
C) Write short note on Optical metallurgical microscope.	CO4	06
Q. 5 Solve Any Two of the following.		
A) What is Refinement of grain size?	CO5	06
B) Write short note on Solid solution strengthening	CO5	06
C) Write the procedure of Dye Penetrant inspection test.	CO5	06

***** End *****

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculator is allowed.
3. Figure to right indicates full marks.
4. Assume suitable data wherever necessary & mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define the following terms i. Unit cell ii. Atomic packing factor iii. Co-ordination number	Remember	6
B) Sketch the following crystal imperfections i. Vacancy ii. Interstitialcy iii substitution impurity	Remember	6
C) Draw stress strain diagram for mild steel, Show effect of carbon content on shape of stress strain diagram.	Understand	6
Q. 2 Solve Any Two of the following.		12
A) What is Solid Solution? Differentiate between substitutional and interstitial solid solution?	Remember	6
B) Draw neatly iron carbon-carbon equilibrium diagram and mention the three variant reactions involved in the phase diagram.	Understand	6
C) What is T-T-T diagram? How it is different from phase diagram? Explain.	Understand	6
Q. 3 Solve Any Two of the following.		12
A) What is annealing? List different types of annealing along with their purpose.	Remember	6
B) Explain with neat sketches different types of flame hardening.	Remember	6
C) Write short notes on the following with their objectives. i. Normalizing ii. Tempering	Understand	6
Q. 4 Solve Any Two of the following.		12
A) Explain steps in specimen preparation for microscopy.	Remember	6
B) Explain the principle of working of optical metallurgical microscope. Compare it with electron microscope.	Understand	6
C) For spark test draw the sparks for the following specimen. i CI ii MS iii HSS	Remember	6
Q. 5 Solve Any Two of the following.		12
A) Write a note on strain hardening.	Understand	6
B) Explain Dye penetrant test. What are its Applications?	Remember	6
C) Explain dispersion strengthening w.r.t. basic mechanism, critical factors, advantage, and commercial examples.	Understand	6

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: Mechanical & Allied Engg. Semester: III

Subject Code & Name: BTMES304 Materials Science and Metallurgy

Max Marks: 60

Date:09-01-24

Duration: 3Hrs.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q. 1 Solve *Any Two* of the following.

12

- A)** Sketch the following directions and planes within a cubic unit cell:

6

$[3\bar{1}2]$

$[10\bar{2}]$

$(2\bar{1}1)$

$(3\bar{1}2)$

- B)** Explain, in brief, the slip and twinning mechanisms of plastic deformation with the help of schematic diagrams.

6

- C)** Compare Brinell Hardness Test with Rockwell Hardness Test.

6

Q.2 Solve *Any Two* of the following.

12

- A)** Explain the Hume-Rothery's rules of solid solubility. Show the solidification of pure metals graphically.

6

- B)** What is lever rule? Derive the statement of Lever rule using schematic diagrams.

6

- C)** Discuss the eutectoid transformation on the equilibrium diagram of steels. Draw the microstructure of the eutectoid steel.

6

Q. 3 Solve *Any Two* of the following.

12

- A)** Explain the mechanism of removal of heat during cooling with the help of schematic diagrams/graphs. Give the requirements of an ideal quenchant.

6

- B)** Differentiate clearly between annealing and normalizing heat treatments. Show the temperature range of these treatments on the equilibrium diagram.

6

- C)** Describe the different types of flame hardening processes. What is the principle of induction hardening treatment?

6

Q.4 Solve <i>Any Two</i> of the following.	12
A) Compare macroscopy with microscopy w.r.t. definition, purposes, specimen preparation, merits , and demerits.	6
B) Define etching. Differentiate clearly between the mechanism of etching for single phase and two phase alloys.	6
C) Explain sulphur print test w.r.t. purpose, procedure, and chemical reactions involved.	6

Q. 5 Solve <i>Any Two</i> of the following.	12
A) Explain dispersion strengthening w.r.t. basic mechanism, critical factors, advantage, and commercial examples	6
B) Discuss age/precipitation hardening process w.r.t. basic principle, conditions, and steps.	6
C) Describe magnetic particle inspection method using schematic diagrams.	6

***** End *****

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